

Machine Learning

Decision Trees

FAST

**DISCOVERING
THE FUTURE**

Topics of previous lecture

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics
- ✓ Basic Linear Classifier
- ✓ K-Nearest Neighbours Classifier
- ✓ Naive Bayes Classifier

Topics of today's lecture

- No Free Lunch Theorem
- Decision trees
- Choosing the best split
- Impurity measures
- Decision Tree Learners
- Pruning
- Cross-Validation

Fundamental issue in ML

- Learning theory claims that a machine learning algorithm can generalize well from a finite training set of examples
- Inductive reasoning (inferring general rules) from a limited set of examples, is not logically valid
- Machine learning promises to find rules that are **probably** correct about **most** members of the set they concern

Fundamental issue in ML

- No training instance has exactly the same vector of feature values as the test instance

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- Different methods approach this issue differently:
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 - Distance-based methods assume that similar instances have similar label values

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- No training instance has exactly the same vector of feature values as the test instance
- Different methods approach this issue differently:
 - Naïve Bayes assumes independence of features
 - Distance-based methods assume that similar instances have similar label values
- Which is the best approach?

No Free Lunch Theorem

- One of the fundamental theorems on learning

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No Free Lunch (NFL) theorem

Averaged over all possible data-generating distributions, every classification algorithm has the same error rate when classifying previously unobserved points.

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- One of the fundamental theorems on learning

No Free Lunch (NFL) theorem

Averaged over all possible data-generating distributions, every classification algorithm has the same error rate when classifying previously unobserved points.

- Intuition behind NFL:
 - No learning algorithm works best for every problem

No Free Lunch Theorem

- One of the fundamental theorems on learning

No Free Lunch (NFL) theorem

Averaged over all possible data-generating distributions, every classification algorithm has the same error rate when classifying previously unobserved points.

- Intuition behind NFL:
 - No learning algorithm works best for every problem
 - If training data covers a very small proportion of the instance space, then on the rest of the instance space the ground truth might be anything, unless we make additional assumptions

- For any learning algorithm there exist "bad cases", that is datasets where the algorithm performs as bad or worse than random guessing

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- NFL for optimization:
 - If an algorithm performs better than random guessing on some class of problems then it will perform worse than random guessing on the remaining problems.

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- For any learning algorithm there exist "bad cases", that is datasets where the algorithm performs as bad or worse than random guessing
- NFL for optimization:
 - If an algorithm performs better than random guessing on some class of problems then it will perform worse than random guessing on the remaining problems.
- The goal of ML is not to find a universal learning algorithm
- The goal is to understand what kinds of distributions are relevant to the "real world" and what kinds of ML algorithms perform well on data drawn from those data-generating distributions.

How to overcome NFL?

- Make assumptions about the domain!

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- Make assumptions about the domain!
- We already know that different models make different assumptions

How to overcome NFL?

- Make assumptions about the domain!
- We already know that different models make different assumptions
- What do decision trees assume?

Lenses dataset

Presbyopic	Young	Spectacle prescription	Astigmatic	Tear production rate	Can use contact lenses
No	Yes	Myope	No	Normal	Yes
No	Yes	Myope	Yes	Reduced	No
No	Yes	Hypermetrope	No	Reduced	No
No	Yes	Hypermetrope	No	Normal	Yes
No	No	Myope	No	Reduced	No
No	No	Myope	No	Normal	Yes
No	No	Myope	Yes	Normal	Yes
No	No	Hypermetrope	No	Reduced	No
No	No	Hypermetrope	Yes	Reduced	No
No	No	Hypermetrope	Yes	Normal	No
Yes	No	Myope	No	Normal	No
Yes	No	Hypermetrope	Yes	Normal	No
No	Yes	Myope	No	Reduced	???
No	Yes	Myope	Yes	Normal	???
No	Yes	Hypermetrope	Yes	Reduced	???
No	Yes	Hypermetrope	Yes	Normal	???
No	No	Myope	Yes	Reduced	???
No	No	Hypermetrope	No	Normal	???
Yes	No	Myope	No	Reduced	???
Yes	No	Myope	Yes	Reduced	???
Yes	No	Myope	Yes	Normal	???
Yes	No	Hypermetrope	No	Reduced	???
Yes	No	Hypermetrope	No	Normal	???
Yes	No	Hypermetrope	Yes	Reduced	???

Lenses dataset

Presbyopic	Young	Spectacle prescription	Astigmatic	Tear production rate	Can use contact lenses
No	Yes	Myope	No	Normal	Yes
No	Yes				No
No	Yes				No
No	Yes				Yes
No	No				No
No	No				Yes
No	No				Yes
No	No				No
No	No	Hypermetrope	Yes	Reduced	No
No	No	Hypermetrope	Yes	Normal	No
Yes	No	Myope	No	Normal	No
Yes	No	Hypermetrope	Yes	Normal	No
No	Yes	Myope	No	Reduced	???
No	Yes	Myope	Yes	Normal	???
No	Yes	Hypermetrope	Yes	Reduced	???
No	Yes	Hypermetrope	Yes	Normal	???
No	No	Myope	Yes	Reduced	???
No	No	Hypermetrope	No	Normal	???
Yes	No	Myope	No	Reduced	???
Yes	No	Myope	Yes	Reduced	???
Yes	No	Myope	Yes	Normal	???
Yes	No	Hypermetrope	No	Reduced	???
Yes	No	Hypermetrope	No	Normal	???
Yes	No	Hypermetrope	Yes	Reduced	???

This dataset is about 24 people who want to start using contact lenses and visit the optician (1 row=1 person)

Lenses dataset

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No	Yes	Myope	No	Normal	Yes
No	Yes	Myope	Yes	Reduced	No
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No	Yes	Hypermetrope	No	Normal	Yes
No	No	Myope	No	Reduced	No
No	No	Myope	No	Normal	Yes
No	No	Myope	Yes	Normal	Yes
No	No	Hypermetrope	No	Reduced	No
No	No	Hypermetrope	Yes	Reduced	No
No	No	Hypermetrope	Yes	Normal	No
Yes	No	Myope	No	Normal	No
Yes	No	Myope	No	Normal	No
No	Yes	Myope	No	Normal	???
No	Yes	Myope	No	Normal	???
No	Yes	Myope	No	Normal	???
No	Yes	Myope	No	Normal	???
No	No	Myope	No	Normal	???
No	No	Hypermetrope	No	Normal	???
Yes	No	Myope	No	Reduced	???
Yes	No	Myope	Yes	Reduced	???
Yes	No	Myope	Yes	Normal	???
Yes	No	Hypermetrope	No	Reduced	???
Yes	No	Hypermetrope	No	Normal	???
Yes	No	Hypermetrope	Yes	Reduced	???

Optician gathers information (columns 1-5) and decides if the person can use contact lenses (last column)

Lenses dataset

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No	Yes	Myope	Yes	Reduced	No
No	Yes	Hypermetrope	No	Reduced	No
No	Yes	Hypermetrope	No	Normal	Yes
No	No	Myope	No	Reduced	No
No	No	Myope	No	Normal	Yes
No	No	Myope	Yes	Normal	Yes
No	No	Hypermetrope	No	Reduced	No
No	No	Hypermetrope	Yes	Reduced	No
No	No	Hypermetrope	Yes	Normal	No
Yes	No	Myope	No	Normal	No
Yes	No	Hypermetrope	Yes	Normal	No
No	Yes	Myope	No	Reduced	???
No	Yes	Myope	Yes	Normal	???
No	Yes	Hypermetrope	Yes	Reduced	???
No	Yes	Hypermetrope	Yes	Normal	???
No	No	Myope	Yes	Reduced	???
No	No	Hypermetrope	No	Normal	???
Yes	No	Myope	No	Reduced	???
Yes	No	Myope	Yes	Reduced	???
Yes	No	Myope	Yes	Normal	???
Yes	No	Hypermetrope	No	Reduced	???
Yes	No	Hypermetrope	No	Normal	???
Yes	No	Hypermetrope	Yes	Reduced	???

Lenses dataset

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No	Yes	Hypermetrope	No	Reduced	No
No	Yes	Hypermetrope	No	Normal	Yes
No	No	Myope	No	Reduced	No
No	No	Myope	No	Normal	Yes
No	No	Myope	Yes	Normal	Yes
No	No	Hypermetrope	No	Reduced	No
No	No	Hypermetrope	Yes	Reduced	No
No	No	Hypermetrope	Yes	Normal	No
Yes	No	Myope	No	Normal	No
Yes	No	Hypermetrope	Yes	Normal	No
No	Yes	Hypermetrope	Yes	Normal	???

Lenses dataset

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0
0	1	1	1	0	???

Lenses dataset

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0
0	1	1	1	0	???

What would you predict for the test instance?

Lenses dataset

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0
0	1	1	1	0	???

First let's consider only the last feature (R).

How to predict?

R	L
0	1
1	0
1	0
0	1
1	0
0	1
0	1
1	0
1	0
0	0
0	0
0	0
0	???

How to predict?

R	L
0	1
1	0
1	0
0	1
1	0
0	1
0	1
1	0
1	0
0	0
0	0
0	0
0	???

Distribution

	L=0	L=1	Majority
R=0	3	4	L=1

How to predict?

R	L
0	1
1	0
1	0
0	1
1	0
0	1
0	1
1	0
1	0
0	0
0	0
0	0
0	???

Distribution

	L=0	L=1	Majority
R=0	3	4	L=1
R=1	5	0	L=0

How to predict?

R	L
0	1
1	0
1	0
0	1
1	0
0	1
0	1
1	0
1	0
0	0
0	0
0	0
0	???

Distribution

	L=0	L=1	Majority
R=0	3	4	L=1
R=1	5	0	L=0

We agreed to predict the most frequent class in the training data!

If R=0 then L=1

If R=1 then L=0

How to predict?

R	L
0	1
1	0
1	0
0	1
1	0
0	1
1	0
1	0
0	0
0	0
0	0
0	???

Distribution

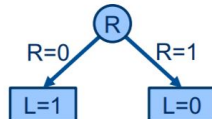
	L=0	L=1	Majority
R=0	3	4	L=1
R=1	5	0	L=0

We agreed to predict the most frequent class in the training data!

If $R=0$ then $L=1$

If $R=1$ then $L=0$

This is called a **decision stump**, that is a decision tree with 1 decision node



How to predict in this case?

A	R	L
0	0	1
1	1	0
0	1	0
0	0	1
0	1	0
0	0	1
1	0	1
0	1	0
1	1	0
1	0	0
0	0	0
1	0	0
1	0	???

How to predict in this case?

A	R	L
0	0	1
1	1	0
0	1	0
0	0	1
0	1	0
0	0	1
1	0	1
0	1	0
1	1	0
1	0	0
0	0	0
1	0	0
1	0	???

Distribution

	L=0	L=1	Majority
A=0, R=0	1	3	L=1
A=0, R=1	3	0	L=0
A=1, R=0	2	1	L=0
A=1, R=1	2	0	L=0

How to predict in this case?

A	R	L
0	0	1
1	1	0
0	1	0
0	0	1
0	1	0
0	0	1
1	0	1
0	1	0
1	1	0
1	0	0
0	0	0
1	0	0
1	0	???

Distribution

	L=0	L=1	Majority
A=0, R=0	1	3	L=1
A=0, R=1	3	0	L=0
A=1, R=0	2	1	L=0
A=1, R=1	2	0	L=0

We agreed to predict the most frequent class in the training data!

If A=0, R=0 then L=1

If A=0, R=1 then L=0

If A=1, R=0 then L=0

If A=1, R=1 then L=0

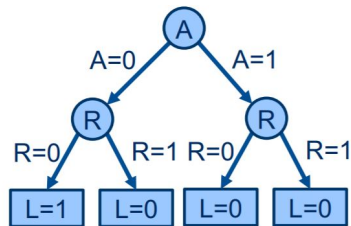
Decision Tree (features A, R)

If $A=0$, $R=0$ then $L=1$

If $A=0$, $R=1$ then $L=0$

If $A=1$, $R=0$ then $L=0$

If $A=1$, $R=1$ then $L=0$



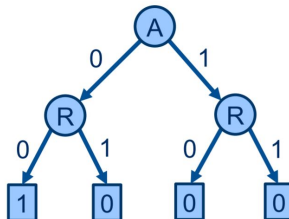
Decision Tree (features A, R)

If $A=0$, $R=0$ then $L=1$

If $A=0$, $R=1$ then $L=0$

If $A=1$, $R=0$ then $L=0$

If $A=1$, $R=1$ then $L=0$



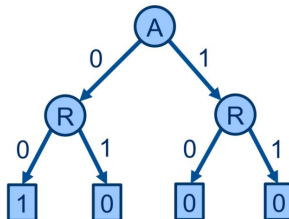
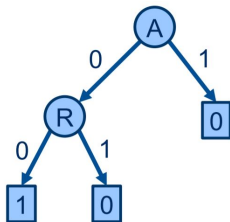
Decision Tree (features A, R)

If $A=0$, $R=0$ then $L=1$

If $A=0$, $R=1$ then $L=0$

If $A=1$, $R=0$ then $L=0$

If $A=1$, $R=1$ then $L=0$



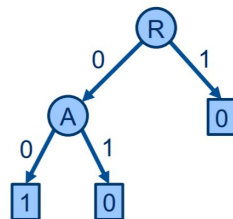
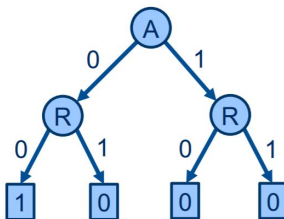
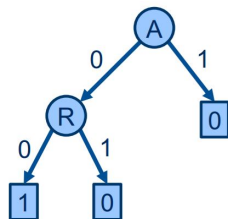
Decision Tree (features A, R)

If $A=0$, $R=0$ then $L=1$

If $A=0$, $R=1$ then $L=0$

If $A=1$, $R=0$ then $L=0$

If $A=1$, $R=1$ then $L=0$



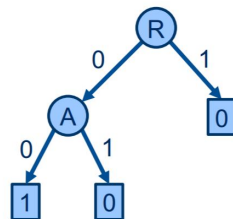
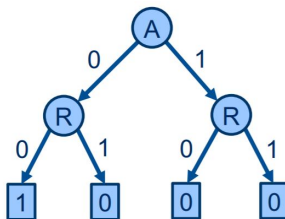
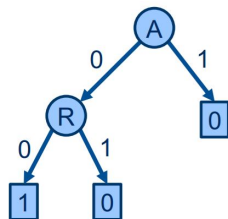
Decision Tree (features A, R)

If $A=0$, $R=0$ then $L=1$

If $A=0$, $R=1$ then $L=0$

If $A=1$, $R=0$ then $L=0$

If $A=1$, $R=1$ then $L=0$

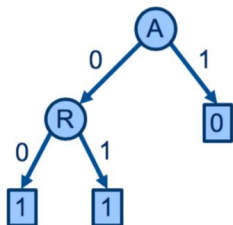


These decision trees are equivalent!

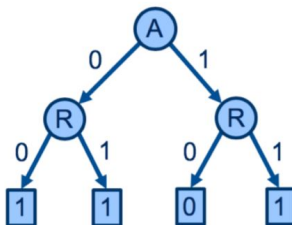
Question

Which of these decision trees are equivalent?

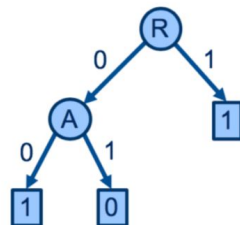
Tree 1



Tree 2



Tree 3



Adding one more feature

H	A	R	L
0	0	0	1
0	1	1	0
1	0	1	0
1	0	0	1
0	0	1	0
0	0	0	1
0	1	0	1
1	0	1	0
1	1	1	0
1	1	0	0
0	0	0	0
1	1	0	0
1	1	0	???

Adding one more feature

H	A	R	L
0	0	0	1
0	1	1	0
1	0	1	0
1	0	0	1
0	0	1	0
0	0	0	1
0	1	0	1
1	0	1	0
1	1	1	0
1	1	0	0
0	0	0	0
1	1	0	0
1	1	0	???

Distribution

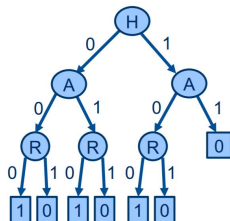
	L=0	L=1	Majority
H=0, A=0, R=0	1	2	L=1
H=0, A=0, R=1	1	0	L=0
H=0, A=1, R=0	0	1	L=1
H=0, A=1, R=1	1	0	L=0
H=1, A=0, R=0	0	1	L=1
H=1, A=0, R=1	2	0	L=0
H=1, A=1, R=0	2	0	L=0
H=1, A=1, R=1	1	0	L=0

Decision Tree (features H, A, R)

	L=0	L=1	Majority
H=0, A=0, R=0	1	2	L=1
H=0, A=0, R=1	1	0	L=0
H=0, A=1, R=0	0	1	L=1
H=0, A=1, R=1	1	0	L=0
H=1, A=0, R=0	0	1	L=1
H=1, A=0, R=1	2	0	L=0
H=1, A=1, R=0	2	0	L=0
H=1, A=1, R=1	1	0	L=0

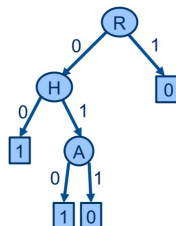
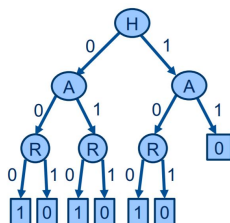
Decision Tree (features H, A, R)

	L=0	L=1	Majority
H=0, A=0, R=0	1	2	L=1
H=0, A=0, R=1	1	0	L=0
H=0, A=1, R=0	0	1	L=1
H=0, A=1, R=1	1	0	L=0
H=1, A=0, R=0	0	1	L=1
H=1, A=0, R=1	2	0	L=0
H=1, A=1, R=0	2	0	L=0
H=1, A=1, R=1	1	0	L=0



Decision Tree (features H, A, R)

	L=0	L=1	Majority
H=0, A=0, R=0	1	2	L=1
H=0, A=0, R=1	1	0	L=0
H=0, A=1, R=0	0	1	L=1
H=0, A=1, R=1	1	0	L=0
H=1, A=0, R=0	0	1	L=1
H=1, A=0, R=1	2	0	L=0
H=1, A=1, R=0	2	0	L=0
H=1, A=1, R=1	1	0	L=0



Decision tree on all features?

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0
0	1	1	1	0	???

Decision tree on all features?

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0
0	1	1	1	0	???

Distribution

	L=0	L=1
...	...	...
P=0, Y=1, H=1, A=0, R=1	1	0
P=0, Y=1, H=1, A=1, R=0	0	0
...	...	...

Decision tree on all features?

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0
0	1	1	1	0	???

Distribution

	L=0	L=1
...	...	...
P=0, Y=1, H=1, A=0, R=1	1	0
P=0, Y=1, H=1, A=1, R=0	0	0
...	...	...

What to predict for this case?

Building a decision tree by recursively splitting the training data

- Let's fix the order of decision nodes to be P, Y, H, A, R

Building a decision tree by recursively splitting the training data

- Let's fix the order of decision nodes to be P, Y, H, A, R

- Split by first feature P:

	L=0	L=1	Pure?
P=0	6	4	No
P=1	2	0	Yes, L=0

Building a decision tree by recursively splitting the training data

- Let's fix the order of decision nodes to be P, Y, H, A, R

- Split by first feature P:

	L=0	L=1	Pure?
P=0	6	4	No
P=1	2	0	Yes, L=0

- P=1 is **pure** (all points belong to the same class), no need to split further

Building a decision tree by recursively splitting the training data

- Let's fix the order of decision nodes to be P, Y, H, A, R

- Split by first feature P:

	L=0	L=1	Pure?
P=0	6	4	No
P=1	2	0	Yes, L=0

- P=1 is **pure** (all points belong to the same class), no need to split further

- Split P=0 by feature Y:

	L=0	L=1	Pure?
P=0, Y=0	4	2	No
P=0, Y=1	2	2	No
P=1	2	0	Yes, L=0

Building a decision tree by recursively splitting the training data

- Let's fix the order of decision nodes to be P, Y, H, A, R

- Split by first feature P:

	L=0	L=1	Pure?
P=0	6	4	No
P=1	2	0	Yes, L=0

- P=1 is **pure** (all points belong to the same class), no need to split further

- Split P=0 by feature Y:

	L=0	L=1	Pure?
P=0, Y=0	4	2	No
P=0, Y=1	2	2	No
P=1	2	0	Yes, L=0

- Two new nodes are not pure, continue splitting

Building a decision tree by recursively splitting the training data

- Let's fix the order of decision nodes to be P, Y, H, A, R

- Split by first feature P:

	L=0	L=1	Pure?
P=0	6	4	No
P=1	2	0	Yes, L=0

- P=1 is **pure** (all points belong to the same class), no need to split further

- Split P=0 by feature Y:

	L=0	L=1	Pure?
P=0, Y=0	4	2	No
P=0, Y=1	2	2	No
P=1	2	0	Yes, L=0

- Two new nodes are not pure, continue splitting
- Split recursively P=0, Y=0 and P=0, Y=1 by H

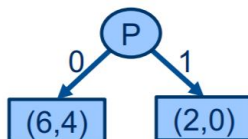
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0

(8,4)

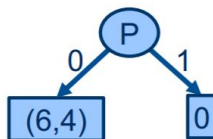
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



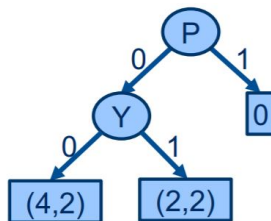
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



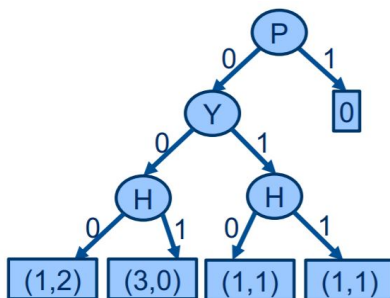
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



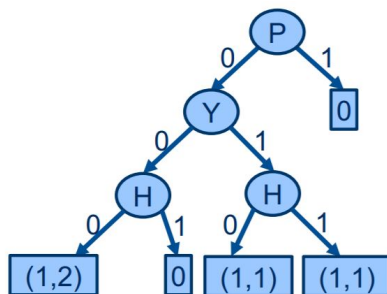
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



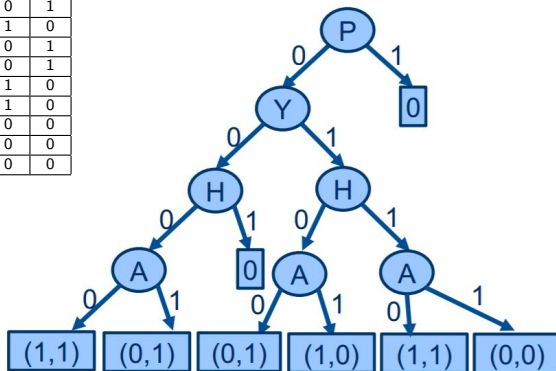
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



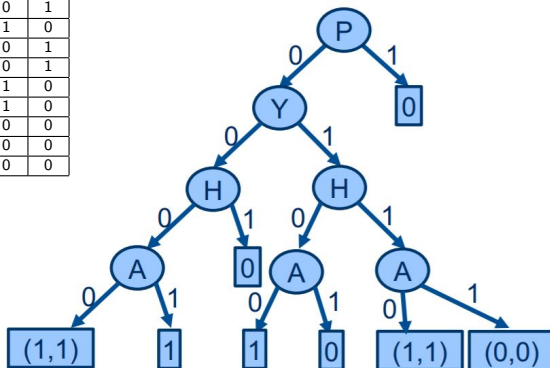
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



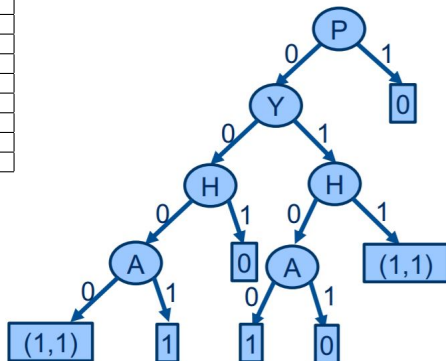
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



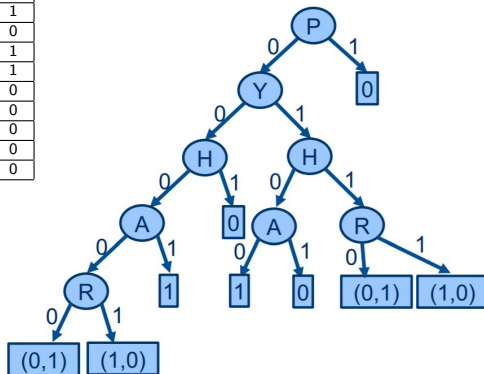
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



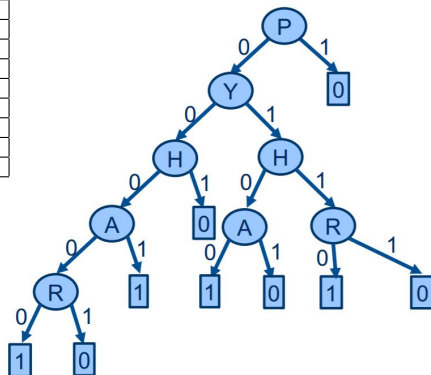
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



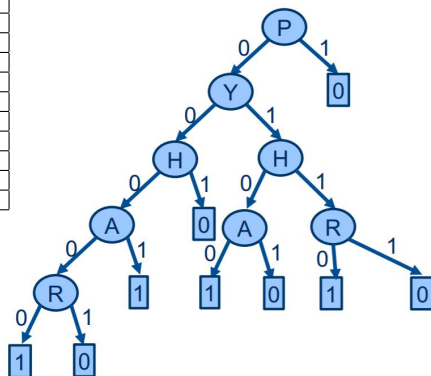
Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Building a decision tree by recursively splitting the training data

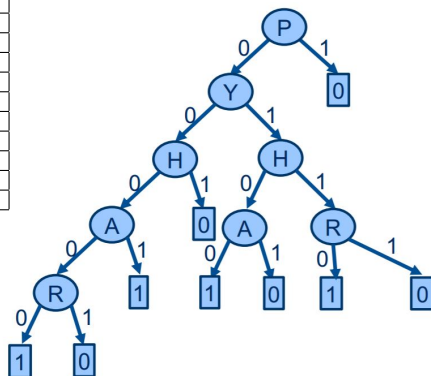
P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Predict for test instance **P=0, Y=1, H=1, A=1, R=0**:

Building a decision tree by recursively splitting the training data

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Predict for test instance $P=0, Y=1, H=1, A=1, R=0$: **L=1**

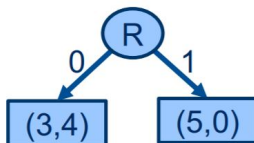
Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0

(8,4)

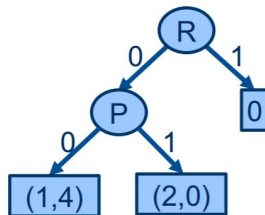
Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



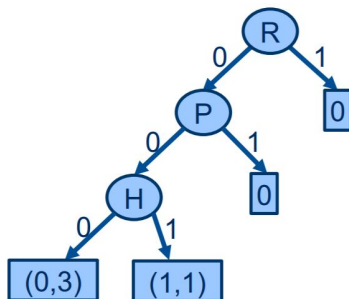
Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



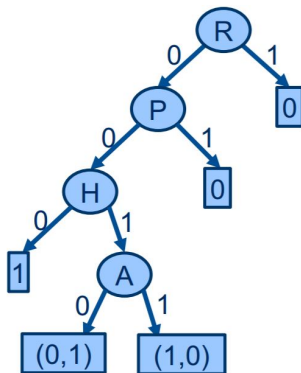
Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



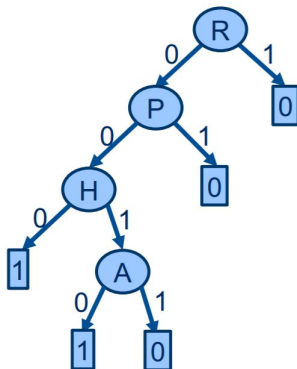
Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



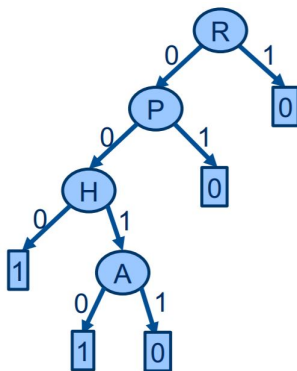
Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

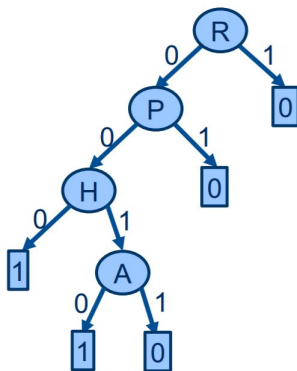
P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Predict for test instance **P=0, Y=1, H=1, A=1, R=0**:

Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

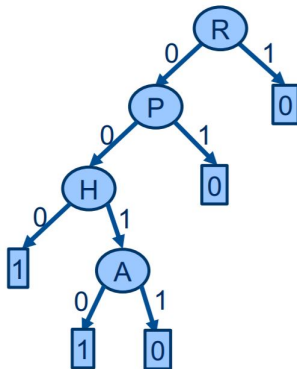
P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Predict for test instance $P=0, Y=1, H=1, A=1, R=0$: **$L=0$**

Building a decision tree by recursively splitting the training data (different fixed order: R, P, H, A, Y)

P	Y	H	A	R	L
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	0	0	1
0	0	0	0	1	0
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	0
0	0	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0



Predict for test instance **P=0, Y=1, H=1, A=1, R=0**: **L=0**

Different order of features resulted in a different tree and prediction

Which decision tree is better?

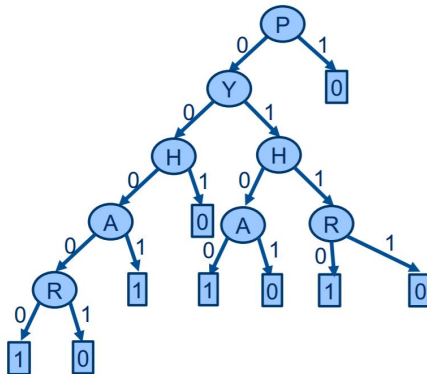
- Consider another test instance: **$P=0$, $Y=0$, $H=0$, $A=1$, $R=1$**

Which decision tree is better?

- Consider another test instance: **$P=0$, $Y=0$, $H=0$, $A=1$, $R=1$**
- What do the 2 trees predict?

Which decision tree is better?

- Consider another test instance: **P=0, Y=0, H=0, A=1, R=1**
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R:

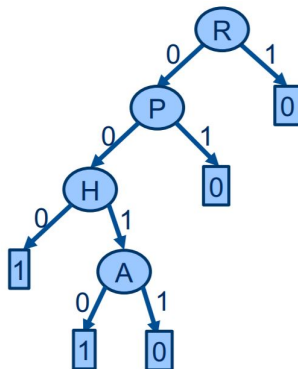


Which decision tree is better?

- Consider another test instance: **P=0, Y=0, H=0, A=1, R=1**
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R: **L=1** (learned from 1 training instance)

Which decision tree is better?

- Consider another test instance: $P=0, Y=0, H=0, A=1, R=1$
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R: $L=1$ (learned from 1 training instance)
 - Decision tree from order R, P, H, A, Y:



Which decision tree is better?

- Consider another test instance: $P=0, Y=0, H=0, A=1, R=1$
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R: $L=1$ (learned from 1 training instance)
 - Decision tree from order R, P, H, A, Y: $L=0$ (learned from 5 training instance)

Which decision tree is better?

- Consider another test instance: $P=0, Y=0, H=0, A=1, R=1$
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R :
"I predict $L=1$ because the only training instance with $P=0, Y=0, H=0, A=1$ (but $R=0$) had $L=1$ "
 - Decision tree from order R, P, H, A, Y :
"I predict $L=0$ because all 5 training instances with $R=1$ always had $L=0$ "

Which decision tree is better?

- Consider another test instance: $P=0, Y=0, H=0, A=1, R=1$
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R:
"I predict $L=1$ because the only training instance with $P=0, Y=0, H=0, A=1$ (but $R=0$) had $L=1$ "
 - Decision tree from order R, P, H, A, Y:
"I predict $L=0$ because all 5 training instances with $R=1$ always had $L=0$ "
- Which one would you believe?

Which decision tree is better?

- Consider another test instance: $P=0$, $Y=0$, $H=0$, $A=1$, $R=1$
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R:
"I predict it is ok for you to use contact lenses because optician said ok to another person who was also presbyopic, myopic and astigmatic except that unlike you he/she had normal tear production rate"
 - Decision tree from order R, P, H, A, Y:
"I predict you should not use contact lenses because optician suggested so to all 5 people with reduced tear production rate"

Which decision tree is better?

- Consider another test instance: $P=0$, $Y=0$, $H=0$, $A=1$, $R=1$
- What do the 2 trees predict?
 - Decision tree from order P, Y, H, A, R:
"I predict it is ok for you to use contact lenses because optician said ok to another person who was also presbyopic, myopic and astigmatic except that unlike you he/she had normal tear production rate"
 - Decision tree from order R, P, H, A, Y:
"I predict you should not use contact lenses because optician suggested so to all 5 people with reduced tear production rate"
- Which one would you believe now?

Which decision tree is better?

- Usually, pure decision nodes covered by more training instances predict better
- Therefore, **usually** smaller trees are better, because on average, they have higher coverage
- Often it is better to stop splitting if the node becomes small (< 10 or < 20 instances), even if the node is not yet pure

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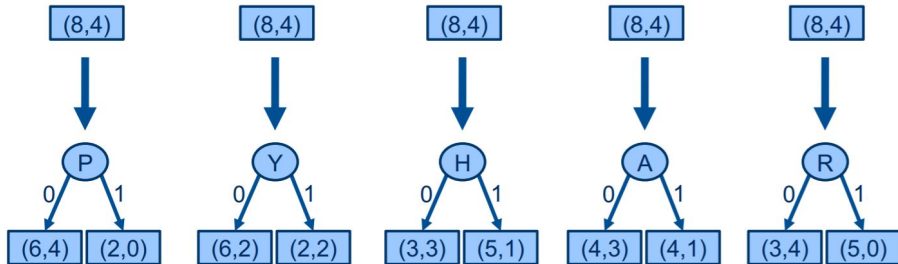
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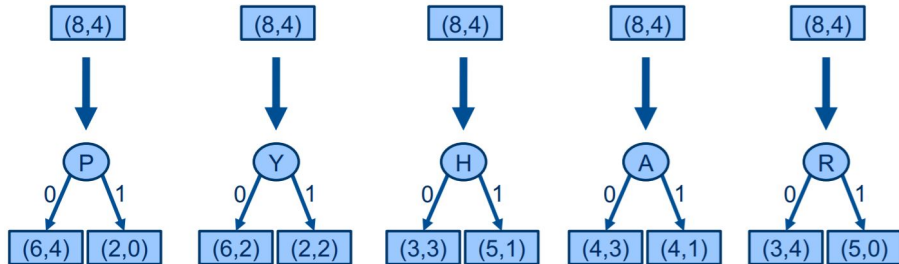
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- Therefore, we need to be able to measure purity

Which split increases purity the most?

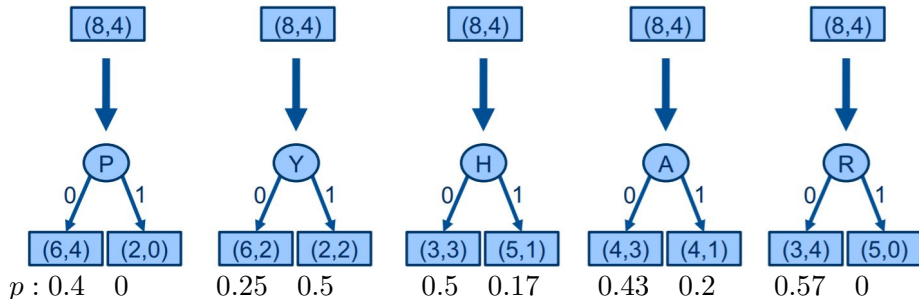


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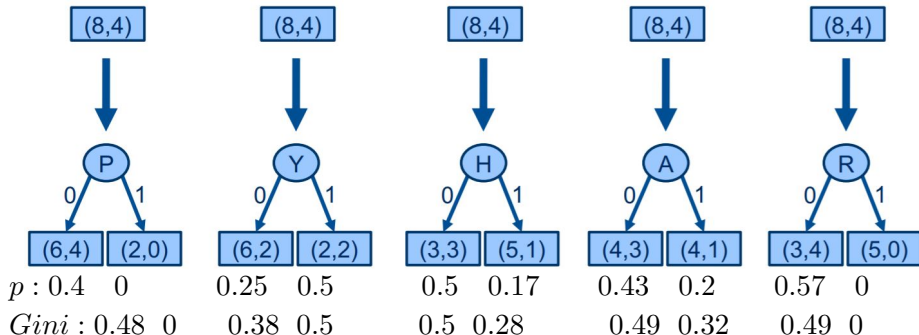
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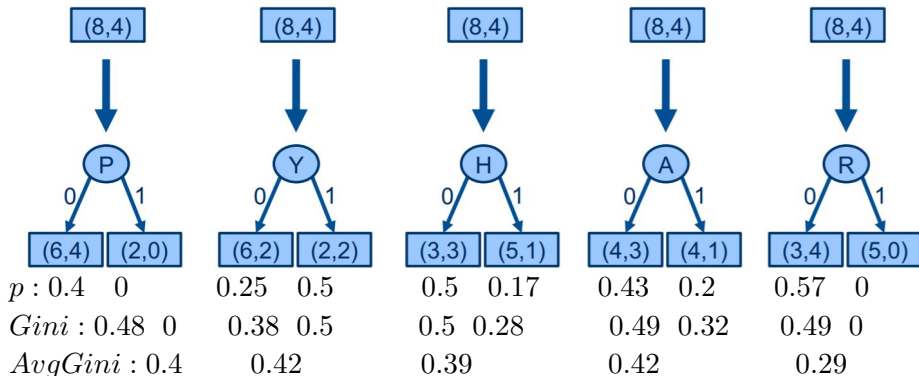
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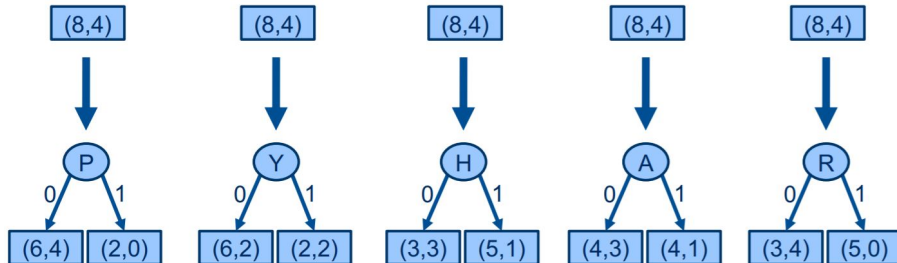


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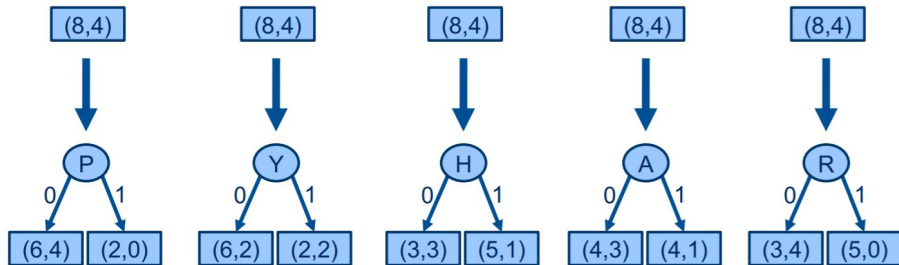
Average Gini impurity after split:

$$AvgGini = \frac{Size_{left}Gini_{left} + Size_{right}Gini_{right}}{Size_{left} + Size_{right}}$$

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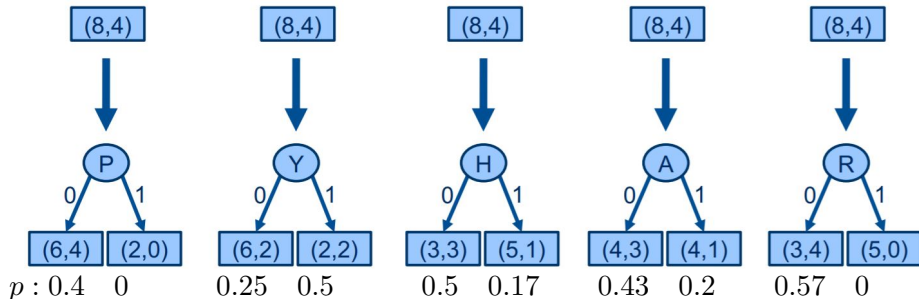


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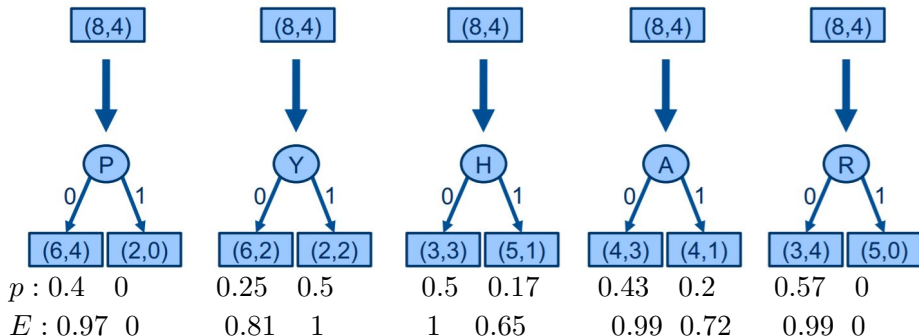
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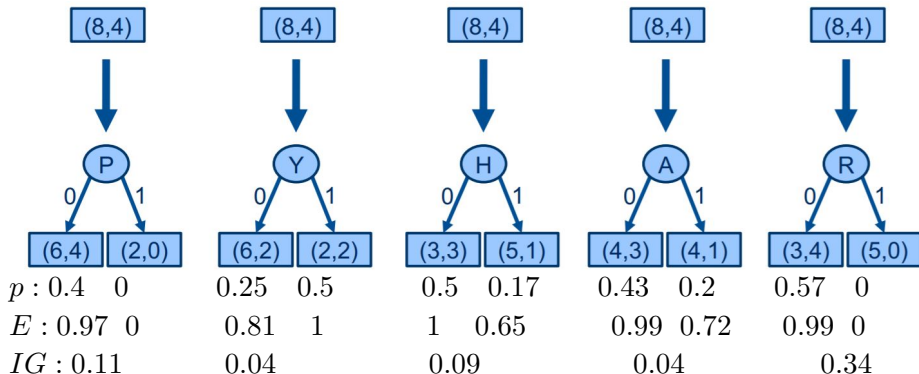
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Entropy of (N_0, N_1) : $E = -p \log_2(p) - (1 - p) \log_2(1 - p)$ where $p = \frac{N_1}{N_0 + N_1}$

Information gain after split: $IG = E_{parent} - \frac{Size_{left}E_{left} + Size_{right}E_{right}}{Size_{left} + Size_{right}}$

Decision Tree Learning

Greedly recursively choosing best splits

Algorithm $\text{GrowDecisionTree}(D, F)$ – grow a decision tree from training data.

Input : data D ; set of features F .

Output : decision tree T with labelled leaves.

```
1 if  $\text{Pure}(D)$  then return  $\text{Label}(D)$ ;
2  $S \leftarrow \text{BestSplit-Class}(D, F)$ ;           // find best feature to split on
3 split  $D$  into subsets  $D_i$  according to the literals in  $S$ ;
4 for each  $i$  do
5   | if  $D_i \neq \emptyset$  then  $T_i \leftarrow \text{GrowDecisionTree}(D_i, F)$ ;
6   | else  $T_i$  is a leaf labelled with  $\text{Label}(D)$ ;
7 end
8 return a tree whose root is labelled with  $S$  and whose children are  $T_i$ 
```

Choosing the best split

- Which is the best feature to split on?
 - The one which minimizes the size of the tree
 - Cannot know the size without recursively building the tree (this would be computationally very hard)
- Intuitively, the splits which result in class distributions closer to purity potentially lead to smaller trees
- DT learners estimate how much purity improves (or how much impurity is reduced) after the considered split

Finding the best feature to split on

Algorithm *BestSplit-Class*(D, F) – find the best split for a decision tree.

Input : data D ; set of features F .

Output : feature f to split on.

```
1  $I_{\min} \leftarrow 1$ ;  
2 for each  $f \in F$  do  
3   split  $D$  into subsets  $D_1, \dots, D_l$  according to the values  $v_j$  of  $f$ ;  
4   if  $\text{Imp}(\{D_1, \dots, D_l\}) < I_{\min}$  ; // if the split reduces impurity  
5   then  
6      $I_{\min} \leftarrow \text{Imp}(\{D_1, \dots, D_l\})$ ;  
7      $f_{\text{best}} \leftarrow f$ ;  
8   end  
9 end  
10 return  $f_{\text{best}}$ 
```

Why these impurity measures?

- What is the intuition behind these measures?
- Guiding principles:
 - Maximize impurity gain:

$$Imp(D) - Imp(\{D_1, \dots, D_l\})$$

the overall impurity of child nodes should be maximally smaller than the impurity of the parent D

- Impurity of child nodes should depend on individual impurities proportionally to their size:

$$Imp(\{D_1, \dots, D_l\}) = \sum_{j=1}^l \frac{|D_j|}{|D|} Imp(D_j)$$

More guiding principles

- Impurity should be defined through the empirical class distribution in the node: in binary classification through $\hat{p}$:

the proportion of positives, i.e. $\hat{p} = \frac{n^{\oplus}}{n^{\oplus} + n^{\ominus}}$

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- Impurity should be maximal if $\dot{p} = \frac{1}{2}$
- These are among the well known impurity measures:
 - Minority Class
 - Gini Index
 - Entropy

Minority class

- Minority class: $\min(\dot{p}, 1 - \dot{p})$
- Measures the proportion of misclassified instances if the leaf was labelled with the majority class
- Equivalently written as $\frac{1}{2} - |\dot{p} - \frac{1}{2}|$

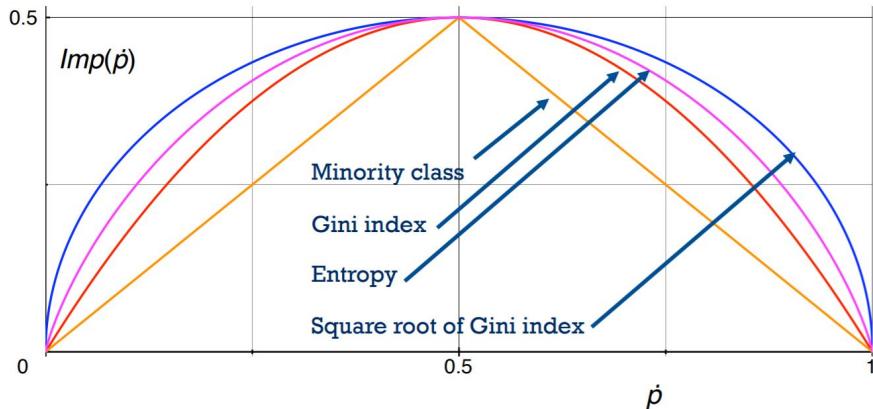
- Gini index: $2\dot{p}(1 - \dot{p})$
- Measures the expected proportion of misclassified instances if the leaf was labelled randomly: positive with probability $\dot{p}$ and negative with probability $1 - \dot{p}$
- The probability of a false positive is then $\dot{p}(1 - \dot{p})$ and the probability of a false negative is $(1 - \dot{p})\dot{p}$
- In case of multiclass classification (there are C classes) the formula is

$$Gini = 1 - \sum_{i=1}^C p_i^2$$

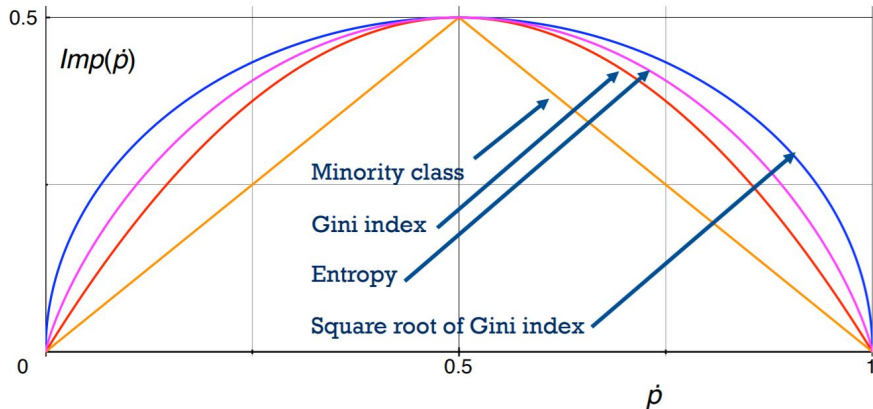
- Entropy: $-\dot{p} \log_2 \dot{p} - (1 - \dot{p}) \log_2 (1 - \dot{p})$
- Measures the expected information, in bits, conveyed by somebody telling you the class of a randomly drawn example
- The purer the set of examples, the more predictable this message becomes and the smaller the expected information
- In case of multiclass classification (there are C classes) the formula is

$$Entropy = - \sum_{i=1}^C p_i \cdot \log_2 p_i$$

Impurity Functions

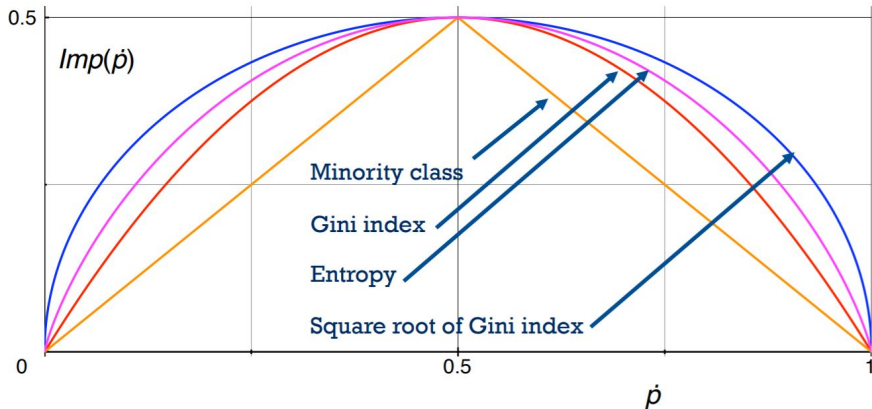


Impurity Functions



What is the best measure?

Impurity Functions



What is the best measure? Depends on the data!

ID3: Iterative Dichotomizer 3

1. Start with a single node
2. Find the feature with the largest information gain
3. Split the node according to this feature
4. Repeat recursively on subnodes

ID3 with categorical non-binary features

- Apply ID3 as usual
- When considering to split using a categorical feature with more than 2 values (non-binary):
 1. Split into k subtrees (instead of just left and right)
 2. Measure entropy in each subtree
 3. Calculate information gain (average entropy per instance after splitting vs before splitting)

ID3 for multi-class classification

- Multi-class classification: Label is categorical, with more than 2 values (classes)
- Apply ID3 as usual
- Formula for entropy for a node with counts $(N_1, N_2, \dots, N_k)$:

$$E = - \sum_{i=1}^k p_i \log_2(p_i),$$

where $p_i = \frac{N_i}{N_1 + N_2 + \dots + N_k}$

C4.5

- Supports continuous attributes
Considers all possible thresholds for splitting, for example $age < 18$; $weight < 100$; $height > 1.95$
- Supports missing values
- Supports pruning
Pruning of a subtree means cancelling all its splits and keeping it as one leaf node

There is also C5 algorithm with some improvements in speed and memory usage.

- Pre-pruning means that during decision tree learning some nodes are not split further
- Decision not to split is typically made if one of the following conditions hold:
 - Node size is below a threshold
 - One of the child nodes would have size below a threshold
 - Depth of the tree is above a threshold
 - Impurity gain is below a threshold

- Post-pruning means that after the tree has been learned, some of the subtrees are removed (replaced by their parent node)
- Decision to prune usually done based on tree performance on a hold-out dataset
- Reduced-error pruning removes subtrees which perform worse than majority class (within the respective parent node)

Pre- or Post-pruning?

- Generally good to do both
- Pre-pruning is faster and does not need any extra data
- Post-pruning is slower, needs extra data, but can "undo" some of the (potentially) bad greedy decisions made at the end of decision tree learning

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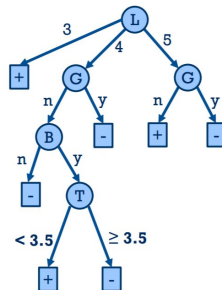
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- Make assumptions about the domain!
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- What do decision trees assume?
- Decision trees assume that the instance space can be split into **segments** such that all (or at least majority of) instances in the segment belong to the same class

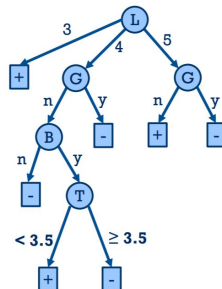
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- What is a segment in decision trees?



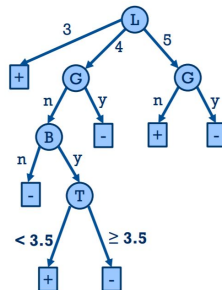
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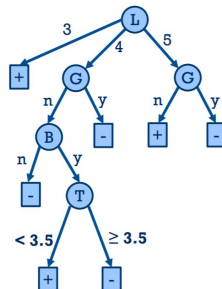
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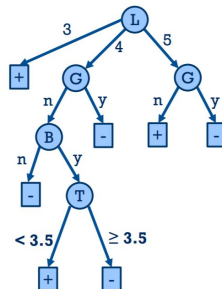
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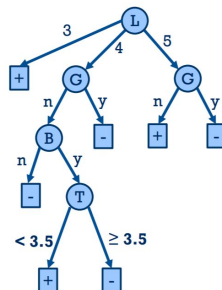
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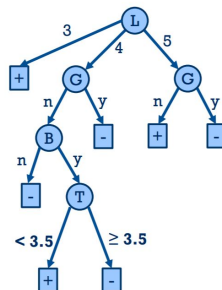
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- Example: $(L = 4) \& (G = y)$



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- Why does a decision tree usually predict better than random?
- Because real-life tasks have usually a structure that favours decision tree learning
- Decision trees have a practically useful inductive bias (set of assumptions about the data) like the other popular learning algorithms

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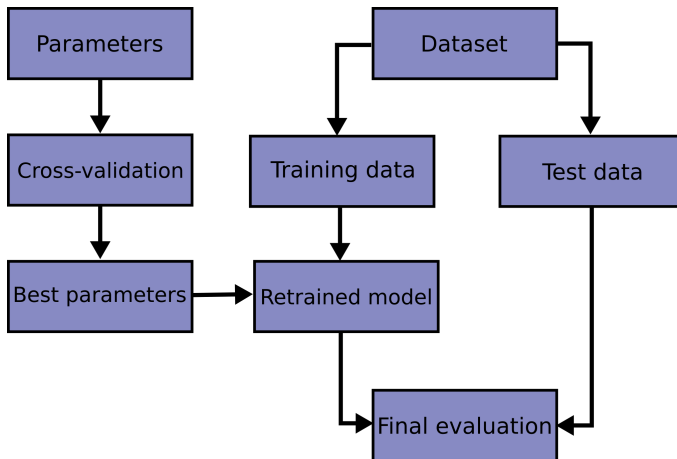
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- The results can depend on a particular random choice for the pair of (train, validation) sets

Choosing hyper-parameter values



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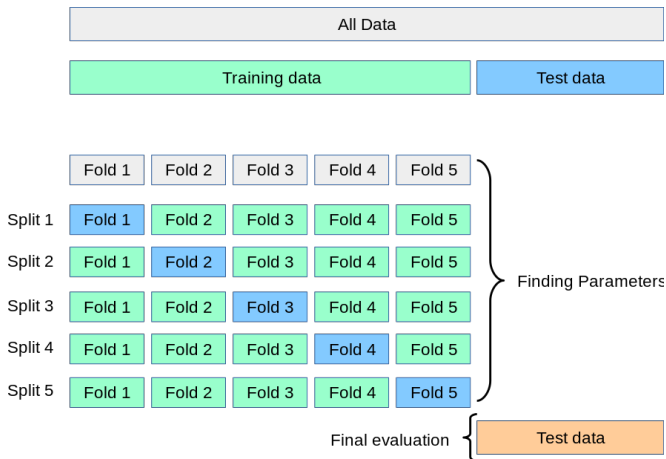
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 - the resulting model is validated on the remaining part of the data
- Average over the k evaluations is considered as a result

K-Fold Cross-Validation



What have we learned today?

- ✓ Kernel Methods
- ✓ No Free Lunch Theorem
- ✓ Decision trees
- ✓ Choosing the best split
- ✓ Impurity measures
- ✓ Decision Tree Learners
- ✓ Pruning
- ✓ Cross-Validation